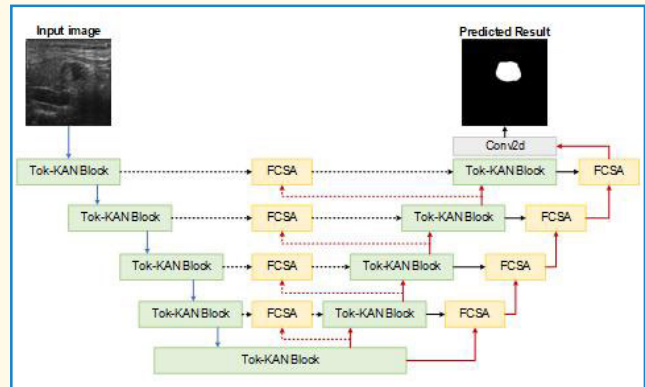


MM-UKAN++: A Novel Kolmogorov–Arnold Network-Based U-Shaped Network for Ultrasound Image Segmentation

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Abstract—Ultrasound (US) imaging is an important and commonly used medical imaging modality. Accurate and fast automatic segmentation of regions of interest (ROIs) in US images is essential for enhancing the efficiency of clinical and robot-assisted diagnosis. However, US images suffer from low contrast, fuzzy boundaries, and significant scale variations in ROIs. Existing convolutional neural network (CNN)-based and transformer-based methods struggle with model efficiency and explainability. To address these challenges, we introduce MM-UKAN++, a novel U-shaped network based on Kolmogorov–Arnold networks (KANs). MM-UKAN++ leverages multilevel KAN layers as the encoder and decoder within the U-network architecture and incorporates an innovative multidimensional attention mechanism to refine skip connections by weighting features from frequency-channel and spatial perspectives. In addition, the network effectively integrates multiscale information, fusing outputs from different scale decoders to generate precise segmentation predictions. MM-UKAN++ achieves higher segmentation accuracy with lower computational cost and outperforms other mainstream methods on several open-source datasets for US image segmentation tasks, including achieving 69.42% IoU, 81.30% Dice, and 3.31 mm HD in the BUSI dataset with 3.17 G floating point of operations (FLOPs) and 9.90 M parameters. The excellent performance on our automatic carotid artery US scanning and diagnostic system further proves the speed and accuracy of MM-UKAN++. Besides, the good performance in other medical image segmentation tasks reveals the promising applications of MM-UKAN++. The code is available on GitHub.

Index Terms—Attention mechanism, deep learning, Kolmogorov–Arnold network (KAN), multiscale feature integration, ultrasound (US) image segmentation.



I. INTRODUCTION

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ULTRASOUND (US) imaging is noninvasive, nonradiative, and convenient, making it widely used in clinical early diagnosis and screening for conditions such as breast cancer [1], thyroid disease [2], and cardiovascular disease [3]. However, traditional US diagnosis heavily depends on the doctor's clinical experience. Computer-aided diagnosis (CADs) can significantly improve diagnostic accuracy and efficiency. Region of interest (ROI) segmentation has always been a key component of CAD [4], [5]. ROI segmentation helps doctors accurately determine the location and status of lesions. With advancements in robotics technology, ROI segmentation provides detailed perceptual information for medical robotic systems, especially image-guided surgical robotic systems, enabling precise diagnosis and treatment [6], [7], [8].

With the advancement of deep learning, convolutional neural networks (CNNs) have achieved success in various

Highlights

- We propose MM-UKAN++, a KAN-based U-network, for fast, accurate US image segmentation, enhancing explainability with multi-scale feature fusion and multi-dimensional attention.
- MM-UKAN++ achieved 69.42% IoU, 81.30% Dice, and 3.31mm HD on the BUSI dataset with 3.17 G FLOPs and 9.90 M Parameters, outperforming mainstream methods in US image segmentation.
- MM-UKAN++ performed very well in the US image segmentation task, revealing the great potential of KANs for image segmentation tasks and confirming the feasibility of deployment on robotic systems.

image-processing tasks due to their powerful local feature extraction capabilities [9]. UNet [10], as a landmark great work, has demonstrated strong effectiveness in medical image segmentation, establishing itself as a foundational architecture for numerous networks. Variants such as UNet++ [11], VNet [12], Attention-UNet [13], and 3-D UNet [14] have also shown good segmentation performance. However, their ability to recover image details and perform long-range modeling still needs improvement. To overcome the inherent limitations of CNNs, transformer and CNN fusion-based methods, such as TransUNet [15] and Swin-UNet [16], effectively deal with global contextual information and long dependencies. However, transformer-based models are more complex with larger parameters, making them prone to overfitting in medical image datasets. The structured state space sequence model (SSM) is more efficient in long-sequence modeling than transformer [17]. U-Mamba [18] and SegMamba [19] have achieved excellent results in image segmentation tasks by leveraging this advantage.

The characteristics of US images, including high noise levels, low contrast, blurred edges, and significant variation in ROI scales, pose great challenges for accurate and efficient segmentation. Although the existing U-shaped segmentation networks have made some progress in US image segmentation, the use of activation functions, multihead attention, or multilayer perceptrons (MLPs) to capture nonlinear relationships remains limited [37], [38]. As a result, these approaches struggle to handle the complex nonlinear features in US images, including blurred anatomical boundaries caused by speckle noise and artifacts, complex and unevenly distributed tissue textures and echo patterns, and multiscale structural information, which limits the segmentation accuracy [39], [40]. Furthermore, due to the design of the network structure and network depth, the internal decision-making process of these methods is difficult to explain, which lacks sufficient explainability, thus failing to meet the stringent requirements of accurate diagnosis and treatment. Recently, Kolmogorov–Arnold networks (KANs) [20] have gained acceptance for their capabilities in nonlinear modeling and enhancing model explainability. Therefore, integrating KANs into U-shaped network structures holds great potential.

To address these challenges and achieve accurate and efficient US image segmentation, this article proposes a U-shaped architecture based on KAN and multiscale fusion (MM-UKAN++). Similar to UNet, we adopt a multilayer

encoder–decoder skip connection architecture, using tokenized KAN blocks instead of convolutional blocks for layer-by-layer feature extraction. This not only facilitates the nonlinear representation of features but also improves the model's explainability. In addition, our designed frequency-channel-spatial attention (FCSA) redefines the skip connection between the encoder and the decoder, enabling the interactive flow of feature information across different layers. Furthermore, different scales of decoder outputs are fused to enrich more detailed features compared with single-scale outputs. With these innovations and improvements, MM-UKAN++ offers superior nonlinear modeling, feature extraction, feature utilization, and explainability. It effectively avoids the problems of inaccurate segmentation caused by noise, blurred edges, and significant ROI scale variations in US image segmentation. Extensive quantitative and qualitative experiments on several US image segmentation benchmark datasets show that MM-UKAN++ surpasses state-of-the-art (SOTA) methods in segmentation accuracy while maintaining lower computational costs. The main contributions of this article are as follows.

- 1) We developed MM-UKAN++, a U-shaped segmentation architecture that combines the advantages of KAN with the multidimensional attention mechanism and multiscale feature fusion, thereby improving accuracy, real-time performance, and explainability in US image segmentation.
- 2) We introduced FCSA to optimize skip connections across different scales, effectively weighting US image details and facilitating feature interaction. This multidimensional attention addresses low contrast and noise challenges in US images. In addition, our multiscale feature fusion structure improves the utilization of features from various layers within the decoder to handle large-scale variations in ROIs.
- 3) We performed qualitative and quantitative evaluations on four public US image segmentation datasets, demonstrating that MM-UKAN++ outperforms SOTA methods. In addition, we deployed MM-UKAN++ in our carotid US automatic scanning and diagnostic system, achieving high-performance real-time segmentation of carotid blood vessels.

The remainder of this article is organized as follows. Section II reviews related work. Section III describes the architecture of MM-UKAN++. Section IV provides the exper-

imental setup and presents a summary and analysis of the comparison, ablation, and additional experiments. Finally, Section V concludes this article and discusses future work.

II. RELATED WORKS

A. CNN-Based Medical Image Segmentation

UNet introduced the U-shaped deep encoder–decoder segmentation architecture, which has been widely used in US segmentation tasks. Derivatives UNet++, UNet+++ [21], and nnUNet [22] enhance UNet through dense skip connections, multiscale fusion, and adaptive parameter adjustment, respectively, thereby improving the accuracy and efficiency of US image segmentation. The introduction of the attention mechanism has further improved network performance. Attention-UNet, DCSAU-Net [23], and MSCA-UNet [24] use various attention mechanisms to enhance the model’s ability to capture and represent features of US images, thus improving segmentation accuracy and robustness. DeepLabv3+ [25] and PSPNet [26] effectively fuse multiscale features using dilated convolution and pyramid pooling, partially addressing the challenge of large variation in ROI scales. ConvNeXt [41] partially compensates for CNNs’ inability to model long-range dependencies by adopting design principles from ViT [27], particularly the use of large convolutional kernels. MedNeXt [42] delivers excellent performance on small datasets by incorporating inverted bottleneck residual blocks, progressively expanding convolutional kernel techniques, and applying composite scaling, building upon ConvNeXt. Overall, CNNs have achieved remarkable results in US image segmentation due to their robust local feature extraction capabilities.

B. Transformer-Based Medical Image Segmentation

ViT [27] divides the input image into fixed-size patches, achieving multilevel feature extraction through encoding and decoding. Furthermore, transformers achieve global information sharing through a unique self-attention mechanism, providing a larger receptive field and easier construction of long-range dependencies, which CNNs lack. However, transformers have a larger number of parameters and require more data for training. Recent studies on the fusion of transformers with CNNs have confirmed that this approach can significantly improve model segmentation performance. TransUNet, Swin-UNet, and BRAU-Net [28] integrate transformer, SwinFormer [29], and Bi-Former [30] into a U-shaped structure to balance network complexity and performance. These models demonstrate excellent performance in US image segmentation and offer new ideas for improving U-shaped segmentation networks. Swin UNETR [43] uses a hierarchical Swin-Transformer encoder with CNN decoding to achieve feature fusion through skip connections. Swin UNETR V2 [44], an enhanced version of Swin UNETR, improves segmentation performance with a larger window size and an optimized self-attention mechanism. H2Former [31] combines the advantages of CNN, transformer, and multiscale channel attention to further improve the network segmentation performance. UNesT [45] efficiently captures global and local features using nested self-attention mechanisms

and skip connections. It demonstrates strong performance in handling complex anatomical structures and multiscale information.

C. Mamba-Based Medical Image Segmentation

The SSM-based Mamba leverages excellent long-sequence modeling capabilities and light computational complexity, significantly reducing training costs compared with transformers and showing greater potential in image segmentation. U-Mamba [18] combines the local feature extraction capabilities of CNNs with the long-sequence modeling of SSM, providing a larger receptive field and improving the model’s ability to handle significant changes in ROI scales. SegMamba [19] excels in 3-D medical image segmentation by designing decoders for multiple Mamba blocks to extract features at different scales and defining skip connections through residual networks to fuse multiscale features.

D. KAN-Based Medical Image Segmentation

Although CNNs, transformers, and MLPs perform well in US image segmentation tasks, they still struggle to effectively model complex nonlinear information in US images. In addition, their internal decision-making process is opaque, often regarded as a black-box operation, and thus these models lack explainability. KAN uses the Kolmogorov–Arnold representation theorem to decompose image features into several simple subfunctions, combining the feature information using nonlinear transformation and weighting. The role of each subfunction is clarified throughout the feature extraction process, effectively extracting the complex nonlinear relationships of the input features, thereby improving the model’s explainability and nonlinear modeling capability. The recently proposed U-KAN [32] incorporates KAN into the U-Net architecture. U-KAN retains the classical multilayer deep encoder–decoder structure of U-Net while introducing “Tok-KAN blocks” in the deep-level representation near the bottleneck. This block’s design enables the KAN operator to be compatible with the existing convolution-based architectures, enhancing the model’s ability to handle nonlinear features and improving overall explainability. In the US image segmentation task, U-KAN shows more advantages in performance. However, this approach does not consider the importance of multiscale information and attention in US segmentation tasks, indicating potential for further improvements in segmentation accuracy. Our proposed MM-UKAN++ uses these advantages of KAN while combining a deep U-shaped encoder–decoder structure, multiple attention mechanisms, and multiscale feature fusion to achieve more accurate and efficient US image segmentation.

III. METHOD

Our proposed segmentation network architecture named MM-UKAN++ is shown in Fig. 1. It uses a U-shaped deep encoder–decoder structure consisting of five parts: encoding part, decoding part, bottleneck, dense skip connections, and multiscale feature fusion. The encoding and decoding parts each consist of five Tok-KAN blocks. Each encoder layer is connected by downsampling, and each decoder is connected

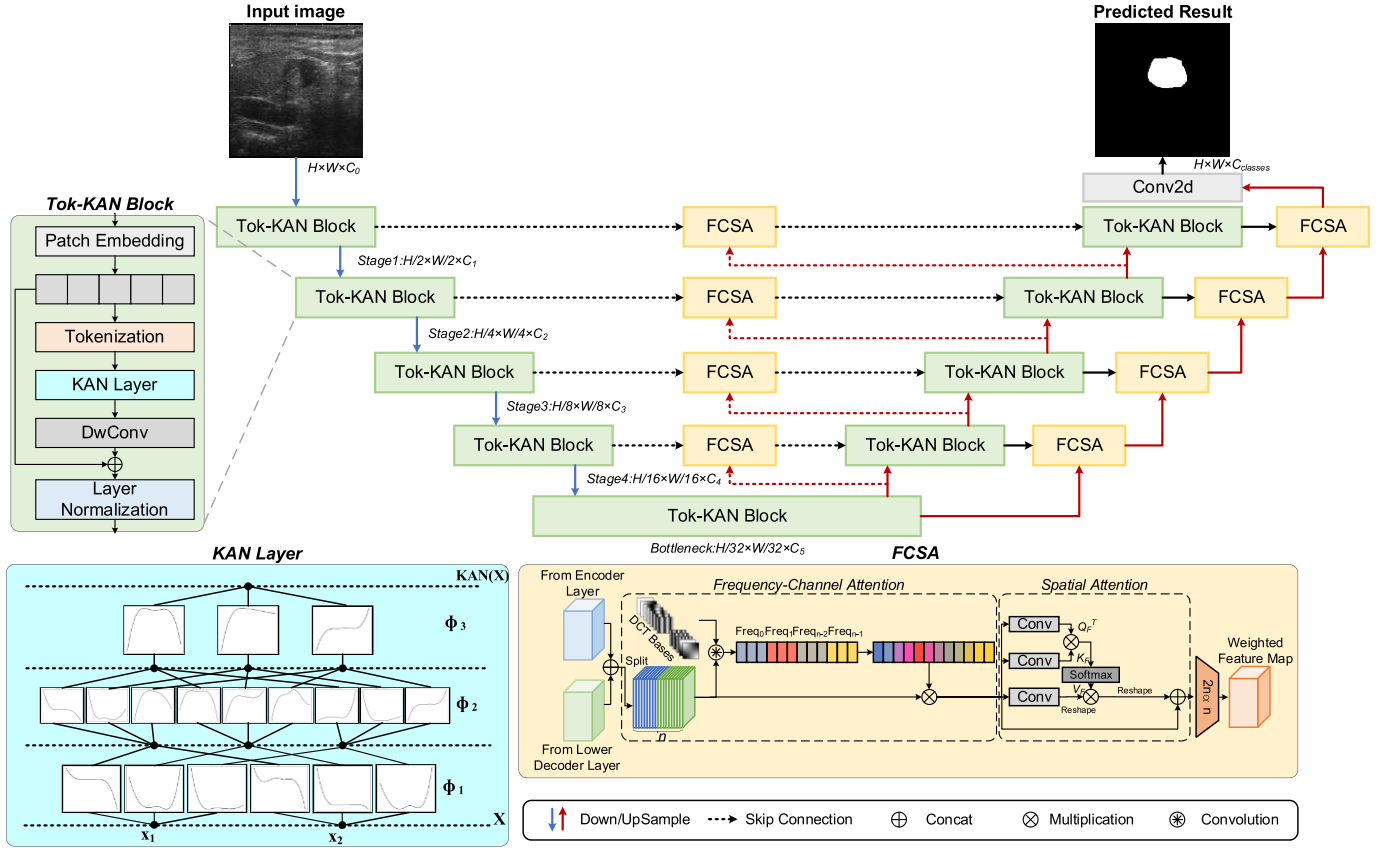


Fig. 1. Overall architecture of the proposed MM-UKAN++. Specific structural details of the Tok-KAN block, KAN layer, and FCSA are also shown in the figure.

by upsampling. FCSA defines dense skip connections for information interaction between different layers of encoder and decoder. Features of different scale output from neighboring decoders are effectively fused with multiscale features through FCSA. The final predicted segmentation result is obtained from the fused feature map via convolution operation.

A. Tok-KAN Block

KANs, based on the Kolmogorov–Arnold representation theorem, have proven to be efficient and interpretable [20]. KANs improve the representation of nonlinear features using spline functions with learnable weights to fit nonlinear functions more accurately. The schematic comparison of MLP and KAN models is shown in Fig. 2.

The expression for the MLP-based neural network is

$$\text{MLP}(X) = (W_{n-1}\sigma \circ W_{n-2}\sigma \circ \dots \circ W_1\sigma \circ W_0)(X) \quad (1)$$

where $X \in \mathbb{R}^{H \times W \times C}$ is the input feature vector, $W_i \in \mathbb{R}^{C_{i+1} \times C_i}$ is the linear transformation matrix, and σ is the nonlinear activation function. In CNNs, feature extraction is achieved through convolutional operations across multiple layers. Both these networks fit complex nonlinear features using multiple layers of nonlinear activation functions. Although effective, these black-box operations do not clarify the specific functions of each internal layer, thereby reducing the model's explainability.

The expression for the KAN of layer n is

$$\text{KAN}(X) = (\Phi_{n-1} \circ \Phi_{n-2} \circ \dots \circ \Phi_1 \circ \Phi_0)(X) \quad (2)$$

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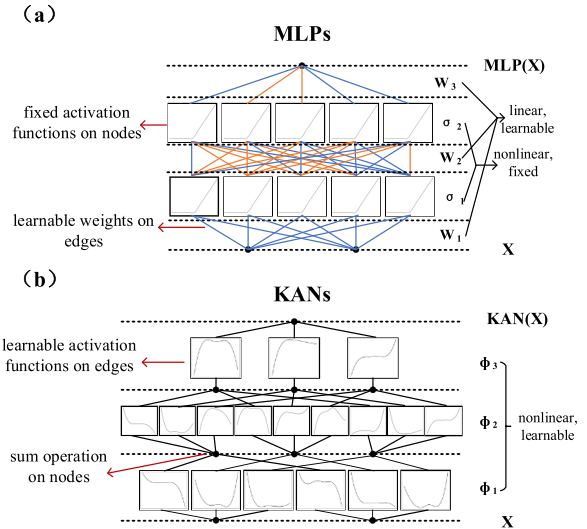


Fig. 2. Comparison of the principles of the MLP model and the KAN model. (a) MLPs: fixed activation and learnable weights. (b) KANs: learnable activation and summation mechanism.

where $\Phi_i \in \mathbb{R}^{n_{i+1} \times n_i}$ is a learnable univariate function matrix, which consists of a function matrix of multiple B-spline functions with the specific expression

$$\Phi_i = \begin{pmatrix} \varphi_{i,1,1} & \varphi_{i,1,2} & \dots & \varphi_{i,1,n_i} \\ \varphi_{i,2,1} & \varphi_{i,2,2} & \dots & \varphi_{i,2,n_i} \\ \vdots & \vdots & \ddots & \vdots \\ \varphi_{i,n_{i+1},1} & \varphi_{i,n_{i+1},2} & \dots & \varphi_{i,n_{i+1},n_i} \end{pmatrix} \quad (3)$$

where $\varphi_{i,a,b} \in \mathbb{R}$ denotes the learnable univariate B-spline function connecting the b th node of the i th layer and the a th node of the $(i + 1)$ th layer, with the expression

$$\varphi_{i,a,b}(x) = \sum_{k=1}^{n_k} c_k B_k(x) \quad (4)$$

where $B_k(x)$ is the B-spline basis function, c_k is the learnable coefficient, and $k \in \{1, 2, \dots, n_k\}$, where n_k represents the number of B-spline basis functions, which is determined by the number of grid points and the spline order. Specifically, in our implementation, we set the number of grid points to 5 and the number of spline order to 3, leading to $n_k = 8$. This configuration ensures a balance between function approximation flexibility and computational efficiency [32]. As shown in (2), KANs learn nonlinear functions through the layer-by-layer combination of multilayer B-spline basis functions. The output of each layer is nonlinearly transformed by a learnable B-spline function matrix Φ_i and passed as input to the next layer. This layer-by-layer B-spline function enables the model to adapt more flexibly to complex features in US images, such as blurred boundaries and noisy regions. Especially when segmenting tissues such as tumors or blood vessels, the model can fit the boundary information accurately at different scales. Besides, the strong local fitting ability of the B-spline function enables the model to capture key features in data of different scales and frequencies without affecting the global feature representation, enhancing the representation of regional variations in US images.

Since each spline function is determinable, the transformation process of each layer and each edge of the model can be explained in terms of an explicit function, thus improving the explainability of the model. In addition, multiple learnable univariate functions replace simple nonlinear activation functions to fit nonlinear features, and these learned univariate functions are also explicit. Therefore, this approach both enhances the ability of the model to represent complex nonlinear features and further improves the explainability of the model. Therefore, we use Tok-KAN blocks, which perform well in UKAN, to represent features [32].

Taking advantage of KAN's strengths in nonlinear modeling and explainability, Tok-KAN first splits the input data into small patches and generates the corresponding embedding vectors $E(x)$. Each embedding vector is then tokenized to obtain a series of tokenized $t_1, t_2, \dots, t_n$. The patch embedding layer uses a convolutional kernel of size 3×3 , stride of 2, and padding of 1. This ensures overlapping patches with sufficient contextual information capture, and the spatial resolution is reduced by half at each stage. These tokenized embeddings serve as the input to the Tok-KAN block, which processes them through the learnable nonlinear transformations to improve feature representation. And the patch embedding parameters for each stage of MM-UKAN++ are shown in Table I. In KAN layer, the model nonlinearly transforms each token by multiple learnable univariate functions $\varphi_{i,a,b}(t_i)$, which allows the complex input features to be better decomposed and facilitates the network to capture more detailed information. Since these univariate functions are determined after network training, these nonlinear transformations are

TABLE I
PATCH EMBEDDING CONFIGURATIONS ACROSS STAGES

Stage	Input Size	Token's Number	Embedded Channels
1	(H, W, C_0)	$(H/2) \cdot (W/2)$	$C_1=32$
2	$(H/2, W/2, C_1)$	$(H/2^2) \cdot (W/2^2)$	$C_2=64$
3	$(H/2^2, W/2^2, C_2)$	$(H/2^3) \cdot (W/2^3)$	$C_3=256$
4	$(H/2^3, W/2^3, C_3)$	$(H/2^4) \cdot (W/2^4)$	$C_4=320$
5	$(H/2^4, W/2^4, C_4)$	$(H/2^5) \cdot (W/2^5)$	$C_5=512$

also explicit. Furthermore, depthwise convolution performs independent in-channel convolution operations for each token, and finally residual concatenation and layer normalization are used after depthwise convolution, which are used to ensure the stability of gradient transfer and avoid model overfitting. The processing of input data by Tok-KAN block can be described as follows:

$$\text{Tok-KAN}(X) = \text{LN}(Z_{k-1}(X) + \text{DwConv}(\text{KAN}(Z_{k-1}(X)))) \quad (5)$$

where $Z_{k-1}(X) \in \mathbb{R}^{((H/2^{k-1}) \cdot (W/2^{k-1})) \times C_{k-1}}$ denotes the tokenized embedding vectors, $\text{DwConv}(\cdot)$ denotes the depthwise convolution operation, $\text{LN}(\cdot)$ denotes the layer normalization operation, and $\text{KAN}(\cdot)$ denotes the three-layer KAN with the expression

$$\text{KAN}(Z_{k-1}) = \Phi_3(\Phi_2(\Phi_1(Z_{k-1}))). \quad (6)$$

Therefore, the Tok-KAN block, built around KAN as the network's core, significantly enhances its ability to model nonlinear information. Its decision-making processes, including tokenization, feature decomposition, and learnable activation functions, are primarily governed by well-defined mathematical structures, ensuring transparency and interpretability. This structured nature arises from the explicit mathematical formulation of the model, wherein each component adheres to well-defined operations, including tokenization for information extraction, feature decomposition for hierarchical representation, and learnable B-spline-based activation functions for nonlinear transformations. However, we acknowledge that certain stochastic factors, such as random initialization and training variations, may introduce slight variability in learned representations. To improve reproducibility and maintain consistency across different training runs, we explicitly set random seeds, ensuring stable parameter initialization, data processing, and computation.

B. Multidimensional Attention Mechanism

We use the multidimensional attention mechanisms to construct dense skip connections between the encoder and the decoder. This enhances the transfer of key features at different scales between different layers, improves the utilization of information at different scales, and reduces the loss of feature information during downsampling. The proposed FCSA is a novel multidimensional attention mechanism that simultaneously weights the input feature maps sequentially in frequency-channel and spatial dimensions. High-frequency information in US images typically corresponds to the boundaries and details of organs and tissues, while low-frequency

information corresponds to their overall morphology and structure. Incorporating frequency information into channel attention allows for better capture of fine image structures and features. Further incorporating spatial attention allows the model to focus on the importance of different spatial locations, enhancing its ability to comprehensively capture both local and global information, and improving the segmentation performance of ROIs at various scales.

1) Frequency-Channel Attention: With the same structure as FcaNet [33], discrete cosine transform (DCT) is first performed on the input feature map $X \in \mathbb{R}^{H_k \times W_k \times C_k}$ ($H_k = H/2^{k-1}$, $W_k = W/2^{k-1}$) to obtain its different frequency components $\{F_1, F_2, \dots, F_n\}$, $n = 16$. Then channel attention is applied to these frequency components, and the attention weights $\beta_i \in \mathbb{R}^{C_k}$ are calculated using global average pooling (GAP) and global maximum pooling (GMP). GAP is used to capture the global contextual information across the entire feature map, while GMP emphasizes the most salient features by focusing on the strongest activations. This combination ensures that both global and key local features are effectively considered in the attention mechanism. The calculation formula is as follows:

$$\beta_i = \text{Sigmoid}(W_1 \cdot \text{GAP}(F_i) + W_2 \cdot \text{GMP}(F_i)) \quad (7)$$

where $F_i \in \mathbb{R}^{H_k \times W_k \times C_k}$ is the frequency component, $\text{Sigmoid}(\cdot)$ is the Sigmoid function, $W_1 \in \mathbb{R}^{C_k \times C_k}$ and $W_2 \in \mathbb{R}^{C_k \times C_k}$ are the learnable weight matrices, and $\text{GAP}(\cdot)$ and $\text{GMP}(\cdot)$ denote the GAP function and the GMP function, respectively. Each frequency component is then weighted by these learnable weights

$$F'_i = \beta_i * F_i \quad (8)$$

where $F'_i \in \mathbb{R}^{H_k \times W_k \times C_k}$ is the weighted frequency component.

Finally, the frequency-channel attention-weighted feature map $X_{FC} \in \mathbb{R}^{H_k \times W_k \times (n \cdot C_k)}$ is

$$X_{FC} = \text{Concat}(F'_1, F'_2, \dots, F'_n) \quad (9)$$

where $\text{Concat}(\cdot)$ is the concatenating operation in the channel dimension.

2) Spatial Attention: The frequency-channel-weighted feature map X_{FC} is initially processed through three distinct convolutions to generate the query Q_F , key K_F , and value V_F ($Q_F, K_F, V_F \in \mathbb{R}^{H_k \times W_k \times (n \cdot C_k)}$). This convolution-based generation method effectively extracts local features while preserving spatial information, thereby enhancing the model's ability to focus on relevant regions in the image [13], [26]. The calculations are as follows:

$$\begin{aligned} Q_F &= \text{Conv}_1(X_{FC}) \\ K_F &= \text{Conv}_2(X_{FC}) \\ V_F &= \text{Conv}_3(X_{FC}). \end{aligned} \quad (10)$$

Then the weight matrix $A_F \in \mathbb{R}^{(H_k \cdot W_k) \times (H_k \cdot W_k)}$ is computed by querying Q_F and key K_F

$$A_F = \text{softmax}(Q_F \cdot K_F^T) \quad (11)$$

where $\text{softmax}(\cdot)$ is the softmax normalization function. The spatial-weighted feature map $X_s \in \mathbb{R}^{H_k \times W_k \times (n \cdot C_k)}$ is obtained by weighting the value vector V_F with the weight matrix A_F

and recovering the spatial dimension by reshape operation, calculated as

$$X_s = \text{reshape}(A_F \cdot V_F) \quad (12)$$

where $\text{reshape}(\cdot)$ is used to adjust the dimensions of X_s . The spatial attention mechanism in FCSA applies Softmax normalization to globally normalize the weights at each spatial location, ensuring that attention weights are interconnected across locations. This mechanism highlights key spatial regions in the image, providing more accurate weight assignment for varying spatial locations and mitigates information loss compared with pooling methods [35].

Finally, the FCSA-weighted feature map $X_{FCSA} \in \mathbb{R}^{H_k \times W_k \times C_k}$ is obtained by applying a residual connection between X_{FC} and X_s , followed by a linear mapping:

$$X_{FCSA} = \text{Linear}(X_s + X_{FC}) \quad (13)$$

where $\text{Linear}(\cdot)$ is used to adjust the channel number to match the input feature map.

C. Multiscale Feature Fusion

High-level features contain more semantic information but lack spatial details, whereas low-level features contain more spatial details but lack semantic information. Therefore, a bottom-up path is used to fuse features of different scales layer-by-layer from the decoder to avoid dilution and loss of information. We use FCSA instead of convolution to fuse features at different scales, making the fused features more comprehensive and improving the model's adaptability to complex backgrounds.

The fusion process involves upsampling the smaller sized feature maps output from the decoder and merging them with the larger sized feature maps from the previous layer into FCSA, resulting in fused feature maps. Assuming the feature maps output from the decoding layer are denoted as $\{C_5, C_4, C_3, C_2, C_1\} \in \mathbb{R}^{H_i \times W_i \times C_i}$, $i = 1, 2, \dots, 5$ in order of size from smallest to largest, the bottom-up fusion process can be represented by the following equation:

$$\begin{cases} C_5^{\text{fuse}} = C_5 \\ C_{i-1}^{\text{fuse}} = \text{FCSA}(C_{i-1}, \text{Upsample}(C_i^{\text{fuse}})) \end{cases} \quad (14)$$

where $C_i^{\text{fuse}} \in \mathbb{R}^{H_i \times W_i \times C_i}$ denotes the fusion results, $\text{FCSA}(\cdot)$ denotes the proposed multidimensional attention mechanism, and $\text{Upsample}(\cdot)$ denotes the upsampling operation. Finally, the fused output of the uppermost layer is converted by a convolution operation to obtain the segmentation prediction result of US image.

D. Loss Function

In US image segmentation, the lesion region usually occupies a small portion of the image, whereas the background covers the majority of pixels. This imbalance causes the model to overoptimize for the majority class, often neglecting the minority class. We propose MM-UKAN++ as an end-to-end segmentation framework. To address class imbalance during model training, we use a combination of Dice loss L_{dice} and cross-entropy loss L_{ce} . L_{ce} aids the model in pixel-level classification, while L_{dice} specifically improves segmentation

TABLE II
DETAILS OF THE DATASET

Datasets	Total	Train	Valid	Test	Standardized Size
BUSI	780	624	78	78	384×384
DDTI	637	447	95	95	256×256
CUBS	2176	1740	218	218	512×512
CAMUS	19232	15386	1923	1923	384×384

performance for minority class regions [12], [22]. The final loss L function is shown in the following equation:

$$L = \lambda_{ce} L_{ce} + \lambda_{dice} L_{dice} \quad (15)$$

where λ_{ce} and λ_{dice} are the weights of L_{ce} and L_{dice} , respectively. Based on our experience and experimental results, we set the weight of λ_{ce} to 0.5 and that of λ_{dice} to 1. This weighting effectively balances global pixel classification while enhancing sensitivity to minority class regions, such as lesions, thereby improving overall segmentation accuracy.

IV. EXPERIMENT

A. Implementation Details

1) *Setup Details*: All the experiments were performed on the same hardware system (NVIDIA GeForce RTX 4090 24G GPU), and all the frameworks were implemented using PyTorch. The networks were trained for 400 epochs using the Adam optimizer, with learning rates optimized by cosine annealing and the batch sizes set to 8. To avoid overfitting and improve model robustness, data augmentation methods such as random flipping and scaling were used. To avoid random chance, each set of experiments was repeated five times.

2) *Evaluation Metrics*: To evaluate the performance of the proposed network comprehensively and fairly, we use the Dice score, intersection over union (IoU), and Hausdorff distance 95th percentile (HD95) as the metrics for segmentation performance. In addition, we use floating point of operations (FLOPs) and the number of parameters to measure the model's complexity.

B. Datasets

Specific details of the datasets we used are shown in Table II. All the datasets were obtained from clinical data sources, where the accuracy and real-time nature of the segmentation results are critical to the diagnostic outcome.

1) *BUSI*: The Breast US Images (BUSI) dataset [46] is a publicly available resource for breast tumor segmentation, comprising 780 breast US images and corresponding segmentation masks, including 133 normal images, 437 benign tumor images, and 210 malignant tumor images. This dataset is widely used in the development and research of CAD systems to promote early detection and diagnosis of breast cancer. The version of the BUSI dataset used in our study was obtained in a preprocessed form provided by the source, where all the images were resized to 384×384 pixels with consistent spacing to ensure data uniformity and facilitate reproducibility.

2) *DDTI*: The Digital Database of Thyroid Images (DDTIs) dataset [48] contains 637 thyroid US images from 290 patients, with each image corresponding to a mask labeled by a professional radiologist. Research on this dataset can help improve early detection and diagnosis of thyroid diseases. The version

of the DDTI dataset used in our study was preprocessed by the source, with all the images standardized to 256×256 pixels and uniform spacing, ensuring consistency across the dataset.

3) *CUBS*: The Carotid US Boundary Study (CUBS) dataset [50] contains 1782 carotid US images from 1088 patients, segmented at the border of the carotid intima media. This dataset is particularly suitable for studying rapid and accurate CAD methods for cardiovascular diseases. The version of the dataset used in this study was preprocessed by the source, where all the images were resized to 512×512 pixels and spacing was standardized, enabling consistent model input and reproducible experiments.

4) *CAMUS*: The Cardiac Acquisitions for Multi-structure Ultrasound Segmentation (CAMUS) dataset [52] was obtained in its original form, where the images have varying sizes and a pixel size of 0.31×0.31 mm. To ensure consistency in input dimensions, we standardized all the images to 384×384 pixels through resampling. And the processed images have the same spacing. This standardization ensures uniformity across the dataset, facilitating consistent training and evaluation of the model.

C. Comparison to SOTA Methods

To validate the effectiveness of MM-UKAN++, we compared it with several SOTA methods on four US datasets. These methods included CNN-based segmentation methods, such as nnUNet (2020), ConvNeXt (2022), and MedNeXt (2023); transformer-based segmentation methods such as TransUNet (2021), Swin UNETR (2021), Swin UNETR-V2 (2023), and UNesT (2023); SSM-based segmentation methods such as U-Mamba (2023); and KAN-based segmentation methods such as UKAN (2024). To ensure objective and fair comparisons, we fully trained each method. Quantitative comparison results for BUSI, DDTI, CUBS, and CAMUS are presented in Tables III–VI. Fig. 3 illustrates the distribution of IoU, Dice coefficients, and HD95 across methods in fivefold cross-validation on the BUSI dataset. As depicted in Fig. 3, MM-UKAN++ achieves the highest median IoU and Dice coefficients, along with the lowest median HD95 among all the methods. In addition, MM-UKAN++ demonstrates minimal performance variation in cross-validation, reflecting its robustness and stability in US image segmentation tasks. These results, along with the quantitative summary in Table III, underscore the significant improvements achieved by the proposed model. The performance trends observed across other datasets are similarly consistent. And the qualitative visual comparisons are shown in Figs. 4 and 5.

As shown in Tables III–V, MM-UKAN++ achieves the highest scores across all three evaluation metrics for all three datasets, with only slightly higher parameters than UKAN. On the BUSI and CUBS datasets, MM-UKAN++ significantly outperforms other models, with improvements of approximately 2.86% in IoU, 3.12% in Dice, and 2.85 mm in HD95 compared with nnUNet. For the DDTI dataset, MM-UKAN++ still outperforms other methods, though the improvement is less pronounced, with IoU at 71.83%, Dice at 83.14%, and HD95 at 3.42 mm. Higher IoU and Dice scores indicate that MM-UKAN++ detects and segments

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT MODELS ON BUSI DATASET

Model	IoU(%)↑	Dice(%)↑	HD95(mm)↓	#Param.(M)↓	FLOPs(G)↓
nnUNet[22]	67.70 $\pm$ 0.29	78.64 $\pm$ 0.28	6.48 $\pm$ 0.68	30.45	26.81
ConvNeXt[41]	64.48 $\pm$ 0.49	75.89 $\pm$ 0.34	9.62 $\pm$ 0.52	28.59	32.49
MedNeXt[42]	65.88 $\pm$ 0.33	76.43 $\pm$ 0.55	8.42 $\pm$ 0.88	60.35	41.57
TransUNet[15]	66.18 $\pm$ 0.27	77.74 $\pm$ 0.52	6.71 $\pm$ 0.69	105.47	111.88
SwinUNETR[43]	67.64 $\pm$ 0.29	78.47 $\pm$ 0.39	6.12 $\pm$ 0.48	62.34	78.40
SwinUNETRv2[44]	67.89 $\pm$ 0.45	79.31 $\pm$ 0.31	5.20 $\pm$ 0.67	75.51	84.13
UNesT[45]	69.04 $\pm$ 0.88	80.84 $\pm$ 0.42	3.42 $\pm$ 0.45	86.24	74.56
U-Mamba[18]	61.81 $\pm$ 0.62	73.42 $\pm$ 0.28	7.45 $\pm$ 0.66	86.30	8.46
UKAN[32]	63.38 $\pm$ 0.59	77.62 $\pm$ 0.29	3.80 $\pm$ 0.52	9.37	14.69
Ours	69.42$\pm$0.24	81.30$\pm$0.33	3.31$\pm$0.48	9.90	7.49

TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT MODELS ON DDTI DATASET

Model	IoU(%)↑	Dice(%)↑	HD95(mm)↓	#Param.(M)↓	FLOPs(G)↓
nnUNet[22]	69.62 $\pm$ 0.49	80.96 $\pm$ 0.36	5.94 $\pm$ 0.81	30.45	12.54
ConvNeXt[41]	68.16 $\pm$ 0.51	78.61 $\pm$ 0.45	8.43 $\pm$ 0.39	28.59	15.42
MedNeXt[42]	69.80 $\pm$ 0.33	79.94 $\pm$ 0.55	6.48 $\pm$ 0.72	60.35	19.45
TransUNet[15]	70.15 $\pm$ 0.64	82.73 $\pm$ 1.32	4.16 $\pm$ 0.46	105.47	52.34
SwinUNETR[43]	70.09 $\pm$ 0.19	82.84 $\pm$ 0.21	7.68 $\pm$ 0.64	62.34	36.72
SwinUNETRv2[44]	70.78 $\pm$ 0.45	83.04 $\pm$ 0.42	6.12 $\pm$ 0.43	75.51	39.36
UNesT[45]	70.96 $\pm$ 0.88	83.03 $\pm$ 0.19	4.92 $\pm$ 0.39	86.24	34.88
U-Mamba[18]	65.91 $\pm$ 0.79	77.54 $\pm$ 1.42	3.98 $\pm$ 0.67	86.30	3.96
UKAN[32]	67.96 $\pm$ 1.13	80.34 $\pm$ 0.94	3.92 $\pm$ 0.34	9.37	6.87
Ours	71.83$\pm$0.65	83.14$\pm$0.48	3.42$\pm$0.59	9.90	3.17

TABLE V
PERFORMANCE COMPARISON OF DIFFERENT MODELS ON CUBS DATASET

Model	IoU(%)↑	Dice(%)↑	HD95(mm)↓	#Param.(M)↓	FLOPs(G)↓
nnUNet[22]	62.89 $\pm$ 0.39	75.91 $\pm$ 0.29	4.68 $\pm$ 0.43	30.45	45.14
ConvNeXt[41]	61.32 $\pm$ 0.42	76.31 $\pm$ 0.43	5.33 $\pm$ 0.86	28.59	55.51
MedNeXt[42]	62.39 $\pm$ 0.55	77.43 $\pm$ 0.56	5.68 $\pm$ 0.96	60.35	70.02
TransUNet[15]	59.48 $\pm$ 0.24	76.67 $\pm$ 0.31	3.64 $\pm$ 0.21	105.47	188.42
SwinUNETR[43]	60.96 $\pm$ 0.68	76.93 $\pm$ 0.42	4.31 $\pm$ 0.23	62.34	132.19
SwinUNETRv2[44]	61.34 $\pm$ 0.43	77.48 $\pm$ 0.19	3.24 $\pm$ 0.61	75.51	141.70
UNesT[45]	65.86 $\pm$ 0.12	78.64 $\pm$ 0.16	2.89 $\pm$ 0.09	86.24	125.57
U-Mamba[18]	56.74 $\pm$ 0.24	71.52 $\pm$ 0.08	3.51 $\pm$ 0.54	86.30	14.26
UKAN[32]	57.61 $\pm$ 0.19	75.69 $\pm$ 0.21	3.22 $\pm$ 0.29	9.37	24.73
Ours	65.92$\pm$0.13	78.72$\pm$0.10	2.68$\pm$0.11	9.90	11.41

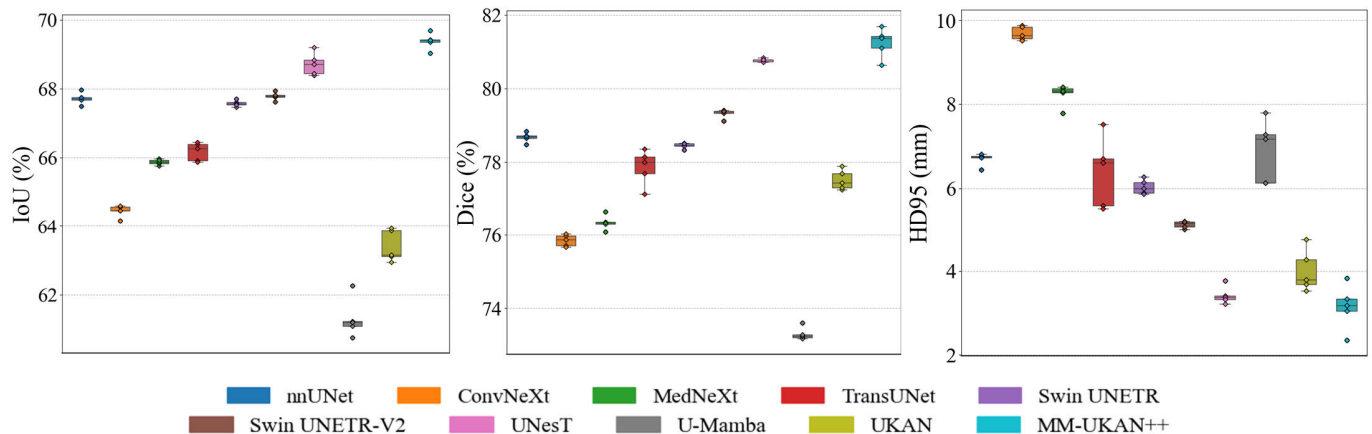


Fig. 3. Boxplot comparison of IoU, Dice, and HD95 for the BUSI dataset.

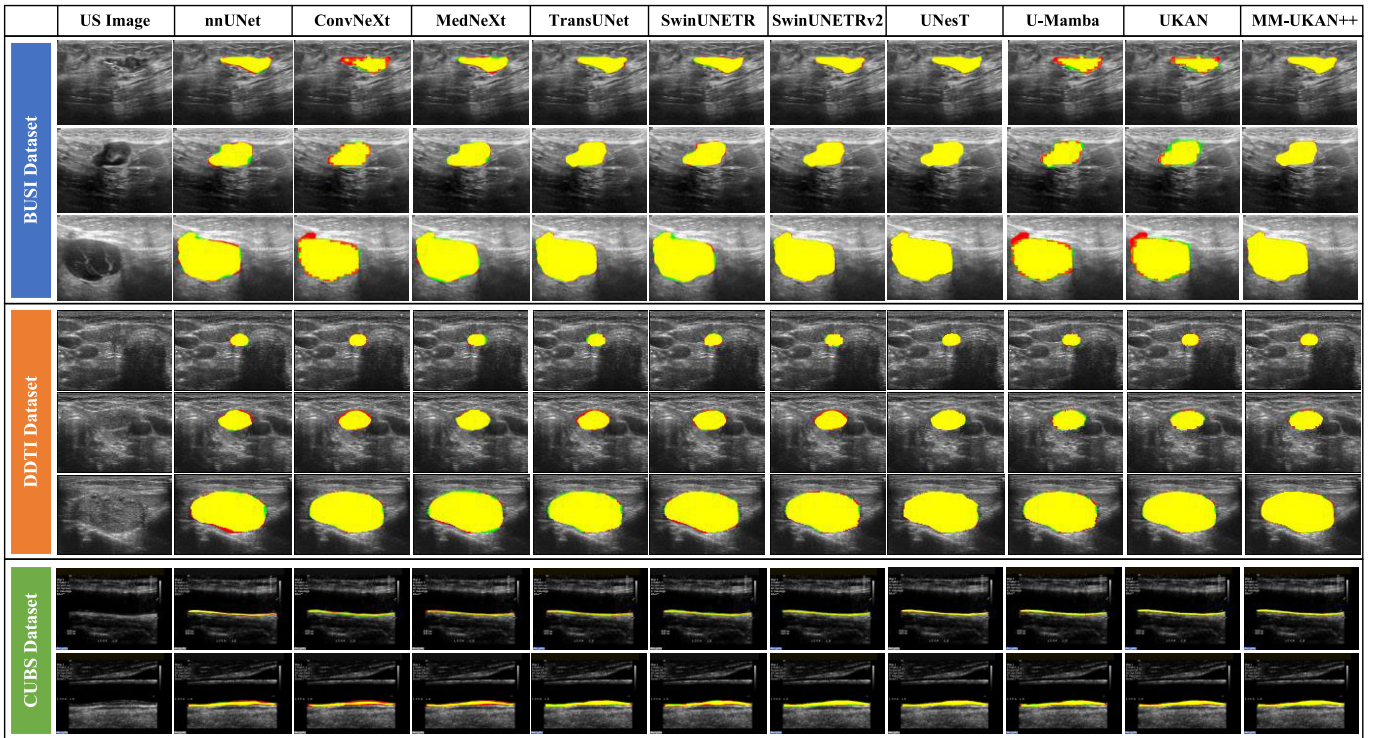


Fig. 4. Visualization of MM-UKAN++ segmentation results with other SOTA methods on BUSI, DDTI, and CUBS. Red areas indicate the label mask. Yellow areas indicate true positives, which are the portions of the predicted region that overlap with the labeled region. Green regions indicate false positives, which are the portions of the predicted region that are not labeled regions.

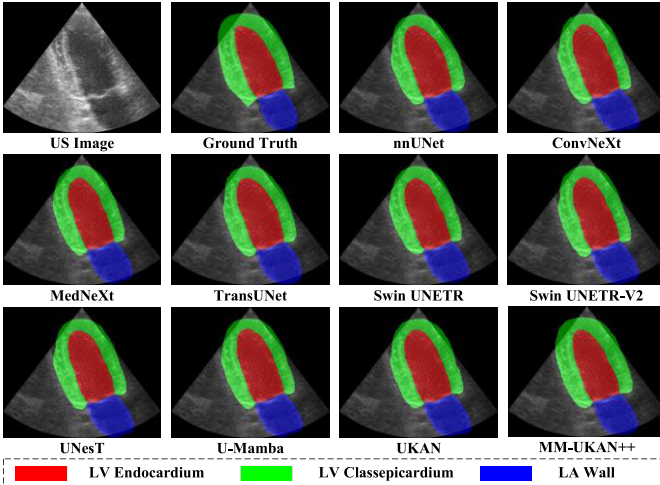


Fig. 5. Visualization of MM-UKAN++ segmentation results with other SOTA methods on CAMUS. Red areas indicate LV endocardium. Green areas indicate LV classepocardium. Blue areas indicate LA wall.

ROIs more accurately with fewer errors. This is because we use the tokenized KAN blocks as the encoder and the decoder, enhancing the model's ability to handle complex nonlinear models in images and significantly reducing network complexity compared to other methods. In addition, we use multiscale feature fusion and dense skip connections to enhance feature fusion and improve target recognition at various scales. Lower HD95 values indicate that MM-UKAN++ provides more precise ROI boundary segmentation. This is due to the multi-attention FCSA design, which optimizes the model's boundary processing capability and reduces boundary blurring, leading to superior HD95 performance for MM-UKAN++. Table VI

presents a detailed comparison of various methods for cardiac multistructural segmentation on the CAMUS dataset, evaluated at end diastole (ED) and end systole (ES). The results indicate that MM-UKAN++ achieves robust segmentation accuracy and precise boundary delineation in both phases. Although Dail-UNet [54] remains the top-performing method in the CAMUS rankings, MM-UKAN++ delivers competitive performance and a distinct advantage over other SOTA methods, including MedNeXt, Swin UNETR V2, and UKAN. While MM-UKAN++ does not surpass Dail-UNet in segmentation accuracy and boundary delineation, its strong performance, combined with an efficient model design, underscores its effectiveness and potential for real-world US image segmentation tasks.

To further evaluate the statistical significance of segmentation performance across different methods, we conducted paired statistical tests on the average performance of the models across four datasets. Fig. 6 presents heatmaps of p -values derived from t -tests, providing a visual comparison of statistical differences between MM-UKAN++ and other mainstream methods. MM-UKAN++ demonstrates statistically significant performance improvements over most other methods, indicating that its advantage over methods such as nnUNet, ConvNeXt, and UKAN is not due to random fluctuations but reflects genuine performance enhancements.

The qualitative comparison results show that MM-UKAN++ has significantly higher recognition accuracy for fine regions, such as intima vascular and tissue boundaries, compared with other methods. When dealing with ROIs of varying sizes and shapes, MM-UKAN++ maintains stable segmentation performance due to its multiscale fusion

TABLE VI
PERFORMANCE COMPARISON OF DIFFERENT MODELS ON CAMUS DATASET

Model	ED			ES			#Param.(M)↓	FLOPs(G)↓
	Mean Dice(%)	Mean HD(mm)	Mean MAD(mm ²)	Mean Dice(%)	Mean HD(mm)	Mean MAD(mm ²)		
Dail-UNet [54]	93.48$\pm$0.44	4.67 $\pm$ 1.91	1.69$\pm$0.63	94.05$\pm$0.53	4.42 $\pm$ 1.24	1.58$\pm$0.54	7.62	15.34
ConvNeXt[41]	91.02 $\pm$ 0.29	5.70 $\pm$ 1.21	2.11 $\pm$ 0.32	91.74 $\pm$ 0.49	5.64 $\pm$ 1.23	2.08 $\pm$ 0.67	28.59	32.49
MedNeXt[42]	91.23 $\pm$ 0.34	5.13 $\pm$ 1.63	1.98 $\pm$ 0.24	92.73 $\pm$ 0.56	5.11 $\pm$ 1.42	1.93 $\pm$ 0.97	60.35	41.57
TransUNet[15]	90.74 $\pm$ 0.45	4.68 $\pm$ 1.03	2.15 $\pm$ 0.51	91.64 $\pm$ 0.21	4.54 $\pm$ 1.15	2.09 $\pm$ 0.35	105.47	111.88
SwinUNETR[43]	91.98 $\pm$ 0.51	4.08 $\pm$ 1.43	1.97 $\pm$ 0.87	93.45 $\pm$ 0.34	3.98 $\pm$ 0.98	1.87 $\pm$ 0.14	62.34	78.40
SwinUNETRv2[44]	92.23 $\pm$ 0.64	3.19 $\pm$ 1.56	1.71 $\pm$ 0.53	93.65 $\pm$ 0.25	3.11 $\pm$ 1.56	1.65 $\pm$ 0.43	75.51	84.13
UNetT[45]	91.92 $\pm$ 0.52	5.06 $\pm$ 1.32	2.34 $\pm$ 0.49	92.84 $\pm$ 0.71	4.96 $\pm$ 1.24	2.29 $\pm$ 0.71	86.24	74.56
U-Mamba[18]	88.26 $\pm$ 0.32	4.68 $\pm$ 2.33	2.17 $\pm$ 0.88	89.64 $\pm$ 0.32	4.59 $\pm$ 1.68	2.11 $\pm$ 0.74	86.30	8.46
UKAN[32]	90.03 $\pm$ 0.56	4.84 $\pm$ 1.43	2.28 $\pm$ 0.52	91.33 $\pm$ 0.18	4.79 $\pm$ 0.97	2.14 $\pm$ 0.86	9.37	14.69
Ours	92.59 $\pm$ 0.28	3.69$\pm$1.02	1.72 $\pm$ 0.63	93.89 $\pm$ 0.68	3.55$\pm$1.45	1.68 $\pm$ 0.56	9.90	7.49

TABLE VII
ABLATION STUDY OF NETWORK'S ARCHITECTURE ON BUSI DATASET. S DENOTES STAGE, AND BNK DENOTES BOTTLENECK

No.	Tok-KAN's Location	IoU(%)↑	Dice(%)↑	HD95(mm)↓	FLOPs (G) ↓	#Parameters (M) ↓
E.1	None	58.32 $\pm$ 0.64	74.86 $\pm$ 0.34	7.24 $\pm$ 0.49	9.89	11.54
E.2	BNK	61.96 $\pm$ 0.91	75.95 $\pm$ 0.26	6.63 $\pm$ 0.62	9.26	11.40
E.3	S4, BNK	64.74 $\pm$ 0.46	78.23 $\pm$ 0.44	4.07 $\pm$ 0.36	7.75	10.51
E.4	S3, S4, BNK	63.64 $\pm$ 0.49	77.60 $\pm$ 0.31	3.99 $\pm$ 0.42	4.90	9.94
E.5	S2, S3, S4, BNK	66.78 $\pm$ 0.18	79.81 $\pm$ 0.26	4.27 $\pm$ 0.16	4.09	9.91
E.6	S1, S2, S3, S4, BNK	69.42$\pm$0.24	81.30$\pm$0.33	3.31$\pm$0.48	3.17	9.90
E.7	S1, S2, S3, S4, BNK (only Encoder)	67.45 $\pm$ 0.26	79.32 $\pm$ 0.44	4.92 $\pm$ 0.54	4.52	10.21

TABLE VIII
ABLATION STUDY OF KAN LAYER DEPTH ON BUSI DATASET

No.	KAN-Layer Depth	IoU(%)↑	Dice(%)↑	HD95(mm)↓	FLOPs (G) ↓	#Parameters (M) ↓
E.8	1	65.78 $\pm$ 0.12	78.02 $\pm$ 0.24	4.55 $\pm$ 0.34	3.11	9.85
E.9	2	67.18 $\pm$ 0.26	79.14 $\pm$ 0.26	3.71 $\pm$ 0.29	3.14	9.88
E.6	3	69.42$\pm$0.24	81.30$\pm$0.33	3.31$\pm$0.48	3.17	9.90
E.10	4	66.85 $\pm$ 0.19	77.65 $\pm$ 0.31	3.84 $\pm$ 0.42	3.20	9.92
E.11	5	67.37 $\pm$ 0.31	80.50 $\pm$ 0.28	3.96 $\pm$ 0.33	3.23	9.94

structure. Furthermore, MM-UKAN++'s accuracy remains high when segmenting targets with significant color differences and morphological changes. The great performance on all four datasets confirms that MM-UKAN++ has outstanding segmentation capability, strong generalization ability, and good consistency in US image segmentation.

D. Ablation Study

We conducted a series of ablation experiments to validate the rationality and effectiveness of MM-UKAN++ in various aspects, including network structure design, KAN layer selection, KAN layer structure, model scale, and the attention mechanism for skip connections. All the experiments were performed on the BUSI dataset using fivefold cross-validation for each group. The specific ablation experiments are as follows.

1) *Effectiveness of Network Structure Design*: We evaluated the segmentation performance of replacing convolutional blocks with Tok-KAN blocks at different stages of the

U-shaped network. The results of the ablation experiments for network structure design are shown in Table VII. It can be seen that using Tok-KAN blocks instead of convolutional blocks in U-shaped networks significantly improves segmentation performance and reduces network complexity. Furthermore, increasing the number of Tok-KAN blocks further enhances network performance due to KAN's lightweight and efficient architecture and superior feature extraction capabilities.

Furthermore, to finalize the network structure, we conducted ablation experiments on an asymmetric model that used CNNs as the decoder. In Experiment E.7, the network's decoder used traditional CNNs instead of Tok-KAN blocks. The quantitative results demonstrate a decrease in all the metrics, proving that Tok-KAN blocks as decoders are better able to handle complex image segmentation tasks compared with CNNs. Therefore, we incorporated Tok-KAN blocks from Stage 1 (S1) to Stage 4 (S4) and the bottleneck of the symmetrical U-shaped network as the final network structure.

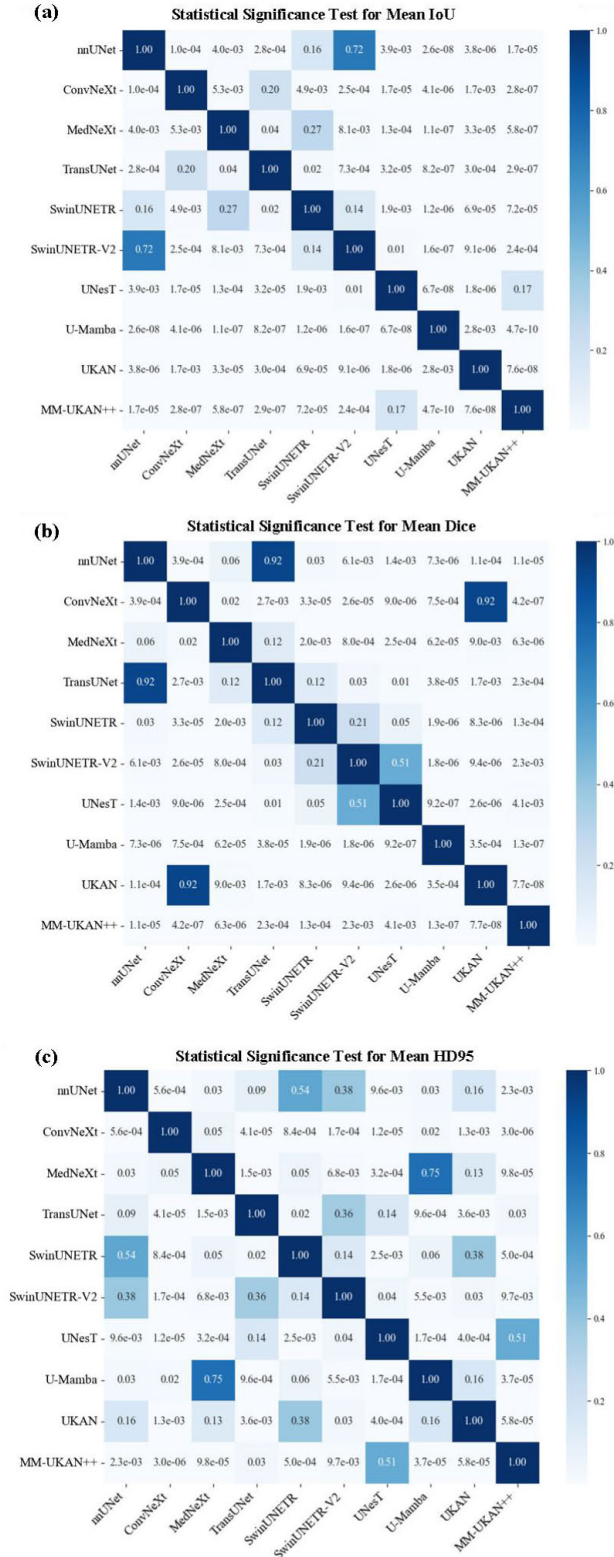


Fig. 6. Heatmaps of statistical significance tests showing the pairwise p -values for average segmentation performance of different methods on four datasets. (a) Statistical significance t-test results for mean IoU. (b) Statistical significance t-test results for mean Dice scores. (c) Statistical significance t-test results for mean HD95. Darker colors represent lower p -values (higher statistical significance), while lighter colors indicate higher p -values (lower statistical significance).

2) *Effectiveness of KAN Layer Depth*: The depth of KAN layer within the Tok-KAN block significantly impacts the

network's performance and complexity. We conducted a series of ablation experiments to determine the optimal depth of KAN layer. The results are shown in Table VIII. Overall, the shallow KAN layer (E.8, E.9) exhibits limited feature extraction ability, while deeper KAN layer (E.6, E.10, E.11) demonstrates stronger feature expression but higher network complexity. However, as the number of layers increases, the network becomes more challenging to optimize, prone to local optima or degradation, leading to performance decline. Therefore, we determined that the optimal depth of KAN layer is 3.

3) *Effectiveness of KAN Utilization*: To verify the advantages of KAN over MLP, we design ablation experiments by constructing KAN layer in Tok-KAN blocks with different combinations of KAN and MLP. The results are presented in Table IX. Replacing one or three KANs with MLPs in the KAN layer significantly decreases the model's segmentation performance and increases its complexity. This indicates that KAN exhibits stronger feature extraction and representation abilities in complex US image segmentation compared with MLP and convolution. These findings further confirm the potential of KAN in medical image segmentation tasks.

4) *Effectiveness of Tok-KAN Tokenizer's Location*: Transformer-based networks typically perform tokenization only once, at the start of the encoding stage. We implement tokenization at different stages of the network to evaluate its performance impact at different positions. The results of the ablation experiments are presented in Table X. Increasing the number of tokenizers at different stages of the network significantly enhances segmentation performance, while only minimally increasing model parameters. Therefore, the optimal results are achieved using tokenizers at all stages, enabling the network to effectively capture multiscale and multilevel features while handling complex patterns and details in the images.

5) *Effectiveness of Tok-KAN's Input Scale*: The number of input channels in the KAN layer of Tok-KAN determines the model size of MM-UKAN++. To establish the optimal model size for the US image segmentation task, we conducted ablation experiments with three different model sizes of MM-UKAN++. The results are shown in Table XI. In comparison to E.20, Tok-KAN in E.6 and E.21 features an increased input scale and a higher number of model parameters, resulting in significant improvements in segmentation performance. However, E.21 exhibits a decrease in Dice and HD95 metrics relative to E.6, despite the increased number of channels and model parameters. To balance segmentation performance and model complexity, we determined the optimal number of input channels for the three layers to be [128, 160, 256].

6) *Effectiveness of Multiscale Fusion*: Multiscale enhances model segmentation performance by fusing feature maps from various network layers, thereby combining coarse-grained global information with fine-grained local features. To evaluate the impact of multiscale fusion, we conducted ablation experiments with different stages of output fusion in the decoder. The experimental results are presented in Table XII. The outputs from all the stages of the fusion decoder demonstrate a significant improvement in segmentation performance, particularly

TABLE IX
ABLATION STUDY OF USING KAN AND USING MLP ON BUSI DATASET

No.	Combination	IoU(%)↑	Dice(%)↑	HD95(mm)↓	FLOPs (G) ↓	#Parameters (M) ↓
E.6	KAN+KAN+KAN	69.42 ± 0.24	81.30 ± 0.33	3.31 ± 0.48	3.17	9.90
E.12	MLP+KAN+KAN	66.70 ± 0.27	79.19 ± 0.34	4.56 ± 0.42	3.44	10.51
E.13	KAN+MLP+KAN	66.59 ± 0.34	79.03 ± 0.16	4.04 ± 0.44	3.44	10.51
E.14	KAN+KAN+MLP	67.01 ± 0.21	79.86 ± 0.26	4.03 ± 0.35	3.44	10.51
E.15	MLP+MLP+MLP	64.14 ± 0.28	77.81 ± 0.29	4.20 ± 0.32	3.98	11.73

TABLE X
ABLATION STUDY OF TOK-KAN TOKENIZER'S LOCATION ON BUSI DATASET

No.	Tokenizer's Location	IoU(%)↑	Dice(%)↑	HD95(mm)↓	FLOPs (G) ↓	#Parameters (M) ↓
E.16	S1	66.78 ± 0.18	79.81 ± 0.26	4.27 ± 0.17	3.08	9.82
E.17	S1, S2	67.25 ± 0.13	80.25 ± 0.45	3.84 ± 0.46	3.10	9.84
E.18	S1, S2, S3	68.05 ± 0.25	80.79 ± 0.54	3.51 ± 0.42	3.12	9.86
E.19	S1, S2, S3, S4	68.94 ± 0.33	81.22 ± 0.16	3.46 ± 0.54	3.13	9.87
E.6	S1, S2, S3, S4, BNK	69.42 ± 0.24	81.30 ± 0.33	3.31 ± 0.48	3.17	9.90

TABLE XI
ABLATION STUDY OF TOK-KAN BLOCKS WITH DIFFERENT SCALES ON BUSI DATASET

No.	Model Size	Input Channels	IoU(%)↑	Dice(%)↑	HD95(mm)↓	FLOPs (G) ↓	#Parameters (M) ↓
E.20	Small	[64,96,128]	65.84 ± 0.16	78.94 ± 0.29	4.39 ± 0.37	0.83	2.50
E.6	Middle	[128,160,256]	69.42 ± 0.24	81.30 ± 0.33	3.31 ± 0.48	3.17	9.90
E.21	Large	[256,320,512]	69.92 ± 0.32	79.93 ± 0.36	4.25 ± 0.26	10.64	21.52

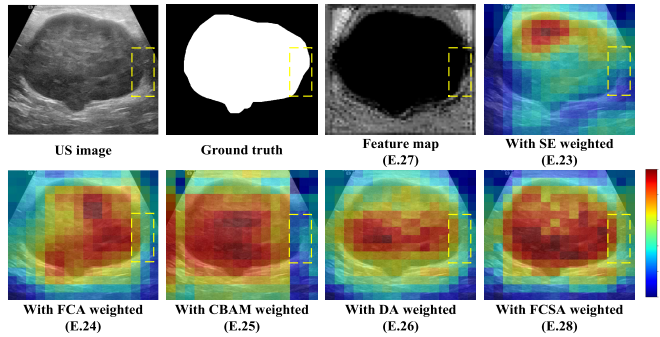


Fig. 7. Visualization results of feature weights obtained by different attention mechanisms.

for IoU and Dice metrics, despite a slight increase in model complexity. Overall, multiscale fusion effectively enhances the model's segmentation performance.

7) Effectiveness of FCSA Design: To validate the superiority of the proposed FCSA in US image segmentation tasks, we tested the effect of MM-UKAN++ using different attention mechanisms, including single attention (SENet [34], FCANet [33]) and hybrid attention (CBAM [35], DANet [36], FCSA). The quantitative results are presented in Table XIII, and the visualization of the feature map weights after applying different attention mechanisms is shown in Fig. 7. Regarding segmentation boundaries, as the marked area in the figure, when comparing the feature maps of the network configuration (E.27) without attention mechanisms, FCANet (E.24) and FCSA (E.28) perform better at extracting edge and texture information in noisy images due to frequency-channel attention, resulting in higher accuracy. They perform well in

boundary segmentation accuracy, as the difference in attention weight values is more pronounced at the edges. In terms of balancing overall and local features, CBAM (E.25), DANet (E.26), and FCSA all incorporate spatial attention, improving segmentation of both large and small regions by capturing local details while maintaining overall consistency. Overall, FCSA enhances tissue boundary and detail extraction through frequency-channel attention, selectively amplifies important channel features via channel attention, and targets key regions of interest with spatial attention. This combination improves segmentation accuracy, robustness, and the recognition of complex structures in US images containing noise.

E. Model Complexity, Convergence, and Explainability

1) Model Complexity: We compared the complexity of MM-UKAN++ with other SOTA methods using parameters and FLOPs. These metrics were calculated based on 256×256 inputs. The comparison results are presented in Table XIV. TransUNet's parameters and FLOPs are 10.65 and 16.51 times higher than MM-UKAN++, respectively, due to its use of the global self-attentive transformer architecture, which demands substantial computational resources. SwinUNETR using a sparse transformer shows improvement but still has significantly higher parameters and FLOPs compared with MM-UKAN++. CNN-based structures such as nnUNet, ConvNeXt, and MedNeXt have slightly higher parameters than MM-UKAN++, but their performances are much more disparate, as shown in Tables III–XI. MM-UKAN++ reduces FLOPs compared with UKAN using more tokenized KANs instead of convolutional blocks, significantly improving seg-

TABLE XII
ABLATION STUDY OF MULTISCALE FUSION ON BUSI DATASET

No.	Fusion Components	IoU(%) $\uparrow$	Dice(%) $\uparrow$	HD95(mm) $\downarrow$	FLOPs (G) $\downarrow$	#Parameters (M) $\downarrow$
E.22	None	65.37 $\pm$ 0.43	78.62 $\pm$ 0.42	4.55 $\pm$ 0.84	3.05	9.81
E.23	S1, S2	66.82 $\pm$ 0.34	79.02 $\pm$ 0.54	4.46 $\pm$ 0.94	3.08	9.83
E.24	S1, S2, S3	68.03 $\pm$ 0.16	79.94 $\pm$ 0.29	4.04 $\pm$ 0.32	3.11	9.85
E.25	S1, S2, S3, S4	68.88 $\pm$ 0.44	80.42 $\pm$ 0.16	3.80 $\pm$ 0.54	3.14	9.88
E.6	S1, S2, S3, S4, BNK	69.42$\pm$0.24	81.30$\pm$0.33	3.31$\pm$0.48	3.17	9.90

TABLE XIII
ABLATION STUDY OF ATTENTION MECHANISMS ON BUSI DATASET

No.	Attention Mechanism Utilized	IoU(%) $\uparrow$	Dice(%) $\uparrow$	HD95(mm) $\downarrow$	#Parameters (M) $\downarrow$
E.23	SE [34]	65.48 $\pm$ 0.32	78.30 $\pm$ 0.16	4.57 $\pm$ 0.22	9.67
E.24	FCA [33]	66.54 $\pm$ 0.16	79.19 $\pm$ 0.42	4.56 $\pm$ 0.38	10.51
E.25	CBAM [35]	66.99 $\pm$ 0.31	79.80 $\pm$ 0.29	4.37 $\pm$ 0.47	9.96
E.26	DA [36]	67.69 $\pm$ 0.21	79.86 $\pm$ 0.26	4.03 $\pm$ 0.35	11.51
E.27	None	65.17 $\pm$ 0.29	77.87 $\pm$ 0.22	3.57 $\pm$ 0.39	9.60
E.6	FCSA	69.42$\pm$0.24	81.30$\pm$0.33	3.31$\pm$0.48	9.90

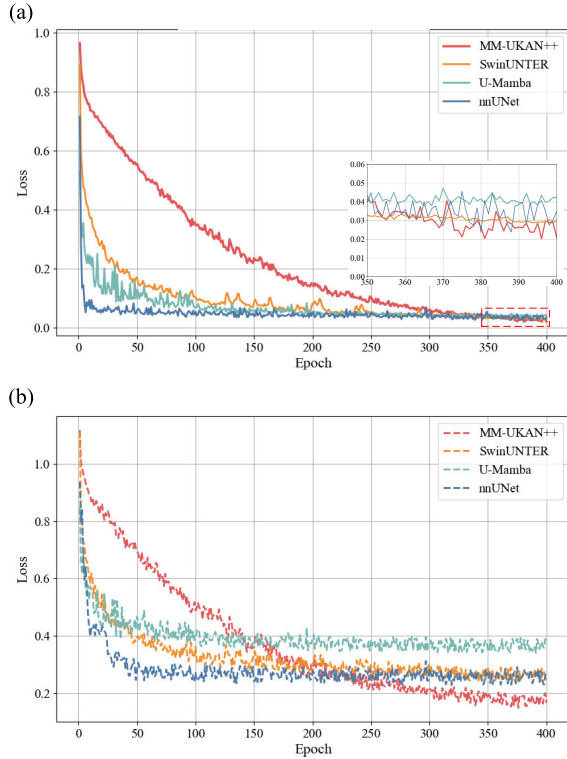


Fig. 8. (a) Training loss curves and (b) validation loss curves of different methods on the BUSI dataset.

mentation performance. Thus, the proposed MM-UKAN++ ensures excellent segmentation results while maintaining a reasonable computational cost, making it a more effective US image segmentation framework that achieves an optimal balance between performance and model complexity.

2) *Model Convergence*: The network's convergence during training, along with its convergence speed, is crucial. We evaluated the convergence of MM-UKAN++ in comparison to other methods over the course of the training process. All the methods were trained for 400 epochs on the

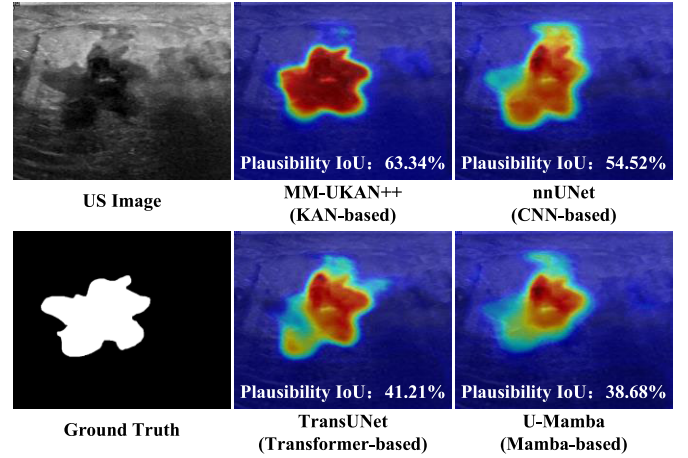


Fig. 9. Visualization results of activation maps for different methods.

BUSI dataset using a consistent training strategy. The loss curves during training and validation are shown in Fig. 8, while training and inference times are provided in Table XV. MM-UKAN++ excels in terms of model parameters, training efficiency, and inference speed; however, its convergence speed is relatively slow. Regarding convergence speed, MM-UKAN++ shows a gradual downward trend. In the initial epochs, the loss decreases at a slower rate. However, in the later stages of training, it demonstrates greater stability, with the loss value stabilizing around 350 epochs and converging with minimal loss. The validation loss curves further illustrate MM-UKAN++'s good generalization ability, with consistently lower validation loss compared with other models. This indicates that MM-UKAN++ not only converges effectively during training but also performs robustly on the validation data, achieving better segmentation performance. MM-UKAN++ requires 10.62 s per epoch for training, significantly less than other models. This highlights MM-UKAN++'s advantage in training efficiency. In terms of inference speed, MM-UKAN++ achieves 34.94 FPS, the

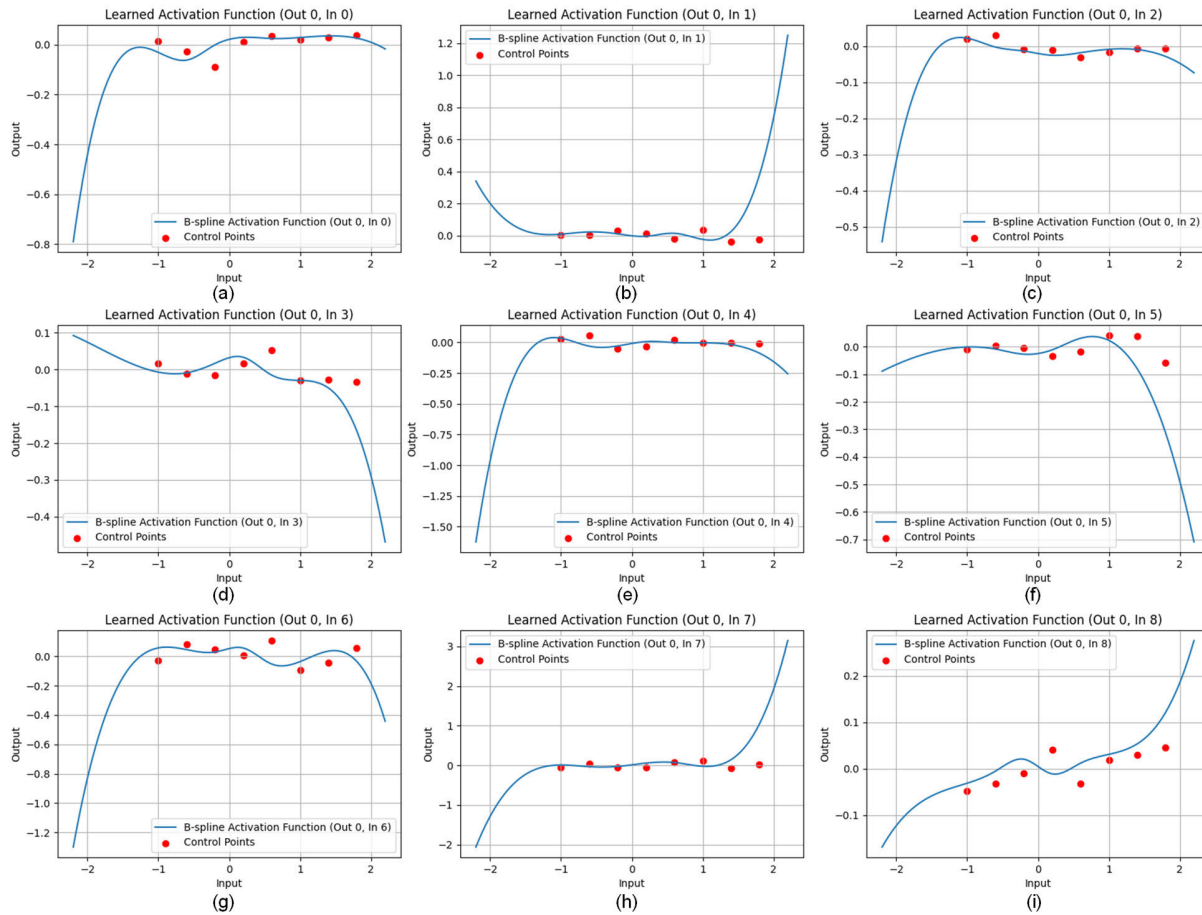


Fig. 10. Visualization of learnable activation functions. (a)–(i) Partial B-spline-based activation functions in the first encoder layer, where the blue curves indicate the activation function curves and the red points indicate the control points.

TABLE XIV

ANALYSIS OF MODEL COMPLEXITY WITH INPUT SIZE OF 256×256

Type	Model	Year	#Parameters(M)↓	FLOPs(G)↓
CNN	nnUNet[22]	2018	30.45	12.54
	ConvNeXt[41]	2022	28.59	15.42
	MedNeXt[42]	2023	60.35	19.45
Trans	TransUNet[15]	2021	105.47	52.34
	SwinUNETR[43]	2021	62.34	36.72
	SwinUNETRv2[44]	2023	75.51	39.36
	UNesT[45]	2023	86.24	34.88
SSM	U-Mamba[18]	2024	86.30	3.96
KAN	UKAN[32]	2024	9.37	6.87
	MM-UKAN++	-	9.90	3.17

highest among all the models, far surpassing Swin UNETR’s 3.57 FPS and U-Mamba’s 4.12 FPS. Model convergence performs similarly on the DDTI, CUBS, and CAMUS datasets. Therefore, the primary advantage of MM-UKAN++ lies in its training stability and efficiency rather than its convergence speed.

3) Model Explainability: To demonstrate the explainability advantages of MM-UKAN++ over other methods, we analyzed the activation values of various methods in activation mode, with the resulting activation maps shown in Fig. 9. The ability of CNN-based nnUNet, transformer-based TransUNet,

TABLE XV

ANALYSIS OF MODEL TRAINING AND INFERENCE PROCESSES

Model	#Parameters (M) ↓	Training time (s/epoch) ↓	Inference time (FPS) ↑
nnUNet[22]	30.45	15.88	20.10
SwinUNETR[43]	62.34	25.42	4.75
U-Mamba[18]	86.30	36.42	5.38
MM-UKAN++	9.90	10.62	45.62

and Mamba-based U-Mamba to recognize activation regions is slightly less accurate than that of KAN-based MM-UKAN++. Based on plausibility IoU [53] comparisons, MM-UKAN++ significantly improves the model’s ability to identify regions of interest and segmentation boundaries. To further demonstrate the strong explainability of MM-UKAN++, we visualized a selection of the learned activation functions from the first layer of its encoder, as depicted in Fig. 10. These visualizations present the B-spline-based activation functions applied between output channel 0 and input channels 0–8 of the first fully connected layer, where the red points indicate the learned control points that define the shapes of these functions. The results illustrate how the encoder’s first layer processes features within US images. Smooth curves represent stable transformations for capturing general patterns, while oscillatory curves reflect sensitivity to finer details and localized

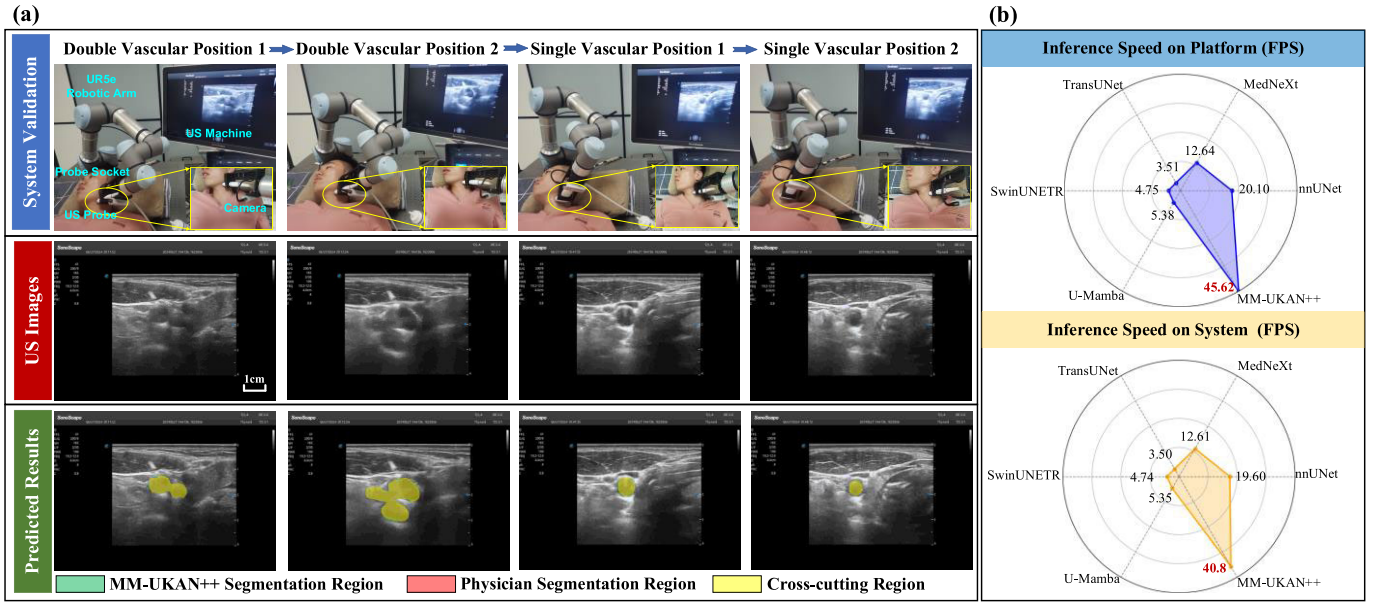


Fig. 11. Additional experimental results. (a) MM-UKAN++ was implemented in our US automatic scanning system for real-time segmentation of carotid arteries. The results were compared with those obtained by professional radiologists. (b) Comparison of model inference speed between MM-UKAN++ and different methods on experimental platforms and automatic scanning diagnostic systems.

variations. This adaptability allows MM-UKAN++ to handle complex nonlinear feature transformations effectively, addressing the unique challenges of US imaging, including low contrast, fuzzy boundaries, and significant variations in ROI scales. The learned activation functions are inherently more flexible and capable of capturing nuanced image characteristics compared with fixed activation functions such as ReLU, which are limited in their ability to model complex relationships in the data. The explicit representation of these learned activation functions enhances the interpretability of the feature transformation process, providing greater transparency into how MM-UKAN++ makes segmentation decisions. By visualizing both the activation maps and the activation functions, we highlight the improved explainability of MM-UKAN++ over traditional segmentation frameworks. This improvement demonstrates not only the model's capability for accurate segmentation but also its ability to provide meaningful insights into its internal decision-making processes.

F. Additional Experiment

Fast and accurate automatic segmentation is crucial for medical robot perception and assisted diagnosis. To verify the feasibility of deployment on a real system, we integrated MM-UKAN++ into our US automatic scanning and diagnostic system [6]. All the experiments involving human US automatic scanning were approved by the Medical Ethics Committee of Weihai Gao District Limin Hospital, Weihai City, Shandong Province, China (2024-02-003). During the scanning process, MM-UKAN++ performs real-time segmentation of carotid arteries for subsequent modeling and analysis. The test results for the segmentation of carotid arterial vessels are shown in Fig. 11. The automatic segmentation results are comparable to the manual segmentation by experienced radiologists. The model achieves an inference speed of 45.62 FPS on the experimental platform. However, since the US imaging device

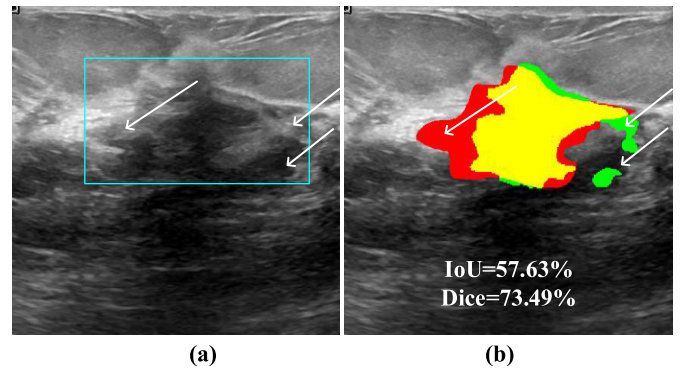


Fig. 12. Failure case in the BUSI dataset. We can see from the segmentation results that there are more incorrect and missing segments. The blue boxes indicate the approximate locations where ROIs are located, and the white arrows indicate where omissions and errors occur. (a) US image. (b) GT and prediction.

operates at 41.00 FPS, the deployed inference speed on the automatic scanning and diagnostic platform is 40.80 FPS. MM-UKAN++ demonstrates a significant improvement over all other methods, indicating its promising application in US automatic scanning and diagnostic systems.

G. Failure Cases Analysis

Analyzing the model's failure cases is meaningful for its optimization. Fig. 12 demonstrates MM-UKAN++ failure cases in segmenting US images. The blue rectangular box indicates that the ROI is highly similar to the surrounding tissues and organs with low contrast, leading to significantly degraded segmentation performance. Incorrect segmentation and boundary anomalies occur at the position marked by the white arrow. This occurs mainly because the features contained inside the segmentation region are more complex, making it difficult for the model to extract accurate information as the feature map size is compressed during downsampling.

To address this problem, a self-attentive mechanism with stronger feature extraction capabilities can be designed. This mechanism would enhance feature extraction and optimize the downsampling process to avoid similar issues.

V. CONCLUSION

In this work, we aim to improve the accuracy and real-time performance of US image segmentation and enhance model explainability by synthesizing multiscale feature fusion and multiple attention mechanisms, resulting in a KAN-based U-shaped network named MM-UKAN++. MM-UKAN++ embeds tokenized KAN blocks instead of convolutional blocks into a U-shaped network to construct the decoder, the encoder, and the bottleneck. This improves the model's ability to represent complex nonlinear patterns in US images, reduces network complexity, and enhances both accuracy and explainability. On several US image segmentation datasets, MM-UKAN++ demonstrates better segmentation results with fewer computational resources compared with previous methods based on CNNs, transformers, and SSM. Significantly, MM-UKAN++ deployed on our US automatic scanning and diagnostic system showed excellent segmentation performance, confirming its promising application. In the future, we plan to further optimize the convergence speed of MM-UKAN++ by improving the structure of ToK-KAN. In addition, we aim to validate MM-UKAN++'s capabilities in other tasks, including US image classification and US-PA (photoacoustic) multimodal image fusion.

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